Curriculum Vitæ

El Mahdi Khribch

Postdoctoral Researcher in Statistics & Machine Learning

Université Paris-Saclay (LMO) · Sorbonne Université (LPSM)

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Research Interests

Monte Carlo methods (importance sampling, MCMC, couplings) · PAC-Bayesian and information-theoretic generalisation bounds · robustness to misspecification · variational inference · Bayesian nonparametrics.

Positions

- **Postdoctoral Researcher** · LMO, Université Paris-Saclay & LPSM, Sorbonne Université — 2026 – present
  ANR project BACKUP (ANR-23-CE40-0018), with Ismaël Castillo, Gilles Blanchard and Guillermo Durand.
- **Visiting PhD Student** · Department of Statistics, University of Warwick — Feb – Mar 2026
- **Research Assistant** · Department of Statistics, Harvard University — 2019

Education

- **PhD in Statistics** · ESSEC Business School & CY Cergy Paris Université (ED 405) — 2022 – 2026
  Thesis — *Contributions to the Theory of Bayesian Computation: Bias, Information, and Robustness.*
  Advisors — Pierre E. Jacob and Pierre Alquier; worked with George Deligiannidis (University of Oxford).
  Committee — Julyan Arbel and Gergely Neu (referees), Stéphanie Allassonnière (chair), Kamélia Daudel.
  Degrees — Docteur en Mathématiques (CY Cergy Paris Université); PhD in Business Administration (ESSEC).
  Funding — ESSEC Foundation doctoral scholarship.
- **M.Sc. Mathematics, Vision & Learning (MVA)** · ENS Paris-Saclay — 2021 – 2022
  Dissertation — *On PAC-Bayesian bounds: faster rates in hostile grounds.*
- **M.Sc. Statistical Science** · University of Oxford — 2020 – 2021
  Dissertation — *On the mixing times of the Metropolized Algorithm with Optimization step (MAO): a new framework.*
- **M.Sc. & Executive Engineering (Diplôme d'ingénieur)** · Mines Paris – PSL University — 2018 – 2022

Teaching

- **High-Dimensional Statistics** · École des Ponts ParisTech (ENPC) — 2023 – 2026
  Teaching assistant (TD/TP), 20 h per year. Lecturer: Badr-Eddine Chérief-Abdellatif.
- **Simulation and Monte Carlo Methods** · ENSAE Paris — 2023 – 2025
  Teaching assistant (TD/TP), 9 h per year, plus 4 h of student project supervision. Lecturer: Nicolas Chopin.
- **Introduction to Stochastic Processes** · ENSAE Paris — 2023 – 2025
  Teaching assistant (TD), 13.5 h per year.

Publications

- **On Importance Sampling and Independent Metropolis–Hastings with an Unbounded Weight Function**
  George Deligiannidis, Pierre E. Jacob, **El Mahdi Khribch**, Guanyang Wang* — *arXiv:2411.09514*
  The Annals of Statistics, to appear

- **Age-Range Dependence and Prediction Compression in Blood-Based Chronological-Age Models**
  Ali Hanbashi, **El Mahdi Khribch**, Eissa A. Jafari and 6 others *doi:10.1186/s12911-026-03898-z*
  BMC Medical Informatics and Decision Making, 2026

* Authors listed in alphabetical order.

Preprints

- **Convex Order Comparisons for Sub-Gamma Random Variables**
  El Mahdi Khribch, Badr-Eddine Chérief-Abdellatif · *Submitted, 2026* *arXiv:2609.02398*
- **Robust Bayesian Inference via Variational Approximations of Generalized ρ-Posteriors**
  El Mahdi Khribch, Pierre Alquier · *Under review, Bernoulli, 2026* *arXiv:2601.07325*
- **Convergence of Statistical Estimators via Mutual Information Bounds**
  El Mahdi Khribch, Pierre Alquier · *Revised and resubmitted, JMLR, 2024* *arXiv:2412.18539*
- **On Mixing Times of the Metropolized Algorithm with Optimization Step (MAO): A New Framework**
  El Mahdi Khribch, George Deligiannidis, Daniel Paulin · *Preprint, 2021* *arXiv:2112.00565*

Working Papers

- **Adaptive Self-Normalized Importance Sampling via Markov Chain Monte Carlo**
  Nicola Branchini, **El Mahdi Khribch**, Víctor Elvira, Gareth O. Roberts · *In preparation*
- **Heavy-Tailed Posterior Modes and Nonparametric Adaptation**
  El Mahdi Khribch, Ismaël Castillo · *In preparation*
- **A PAC-Bayes Theory for f-Divergences: Insights from Convex Ordering**
  El Mahdi Khribch, Badr-Eddine Chérief-Abdellatif · *In preparation*

Awards & Grants

- **Best Poster Prize** · Bayes Comp 2025, Singapore 2025
- **ISBA Travel Grant** · Bayes Comp 2025, Singapore 2025
- **Major-Excellence Scholarship** · French Government, 37,830 €, top 40 students 2018
- **Top 1% Nationwide** · Concours National Commun, Morocco 2018

Talks & Posters

Invited Talks

- **Statify Seminar** · Inria Grenoble Rhône-Alpes Oct 2026
- **FSML Research Group, UCL** · London Mar 2026
- **CSML Reading Group, University of Warwick** · Coventry Feb 2026
- **ESSEC APAC–CNRS–SMU Mini-Workshop** · Singapore Jun 2025
- **BAYSM 2025** · Online Apr 2025

Other Talks

- **Internal Statistics Seminar, LMO** · Université Paris-Saclay, Orsay · *Seminar* Oct 2026
- **MCQMC 2026** · Edinburgh · *Contributed* Jun 2026

Posters

- **ISBA 2026 World Meeting** · Nagoya — Jun 2026
- **Advances in MCMC Methods, EURANDOM** · Eindhoven — Dec 2025
- **Bayes Comp 2025** · Singapore · *Best Poster Prize* — Jun 2025
- **Bayes Comp 2023** · Levi, Finland — Mar 2023

Academic Service

- **Journals** Reviewer for *Statistics and Computing*.
- **Conferences** Reviewer for NeurIPS and AISTATS.

Skills

- **Programming** Python, R, LaTeX, PyTorch, TensorFlow.
- **Languages** English (bilingual), French (native), Arabic (native), Spanish (B1).